The Puzzle of Known Violation

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1. The Puzzle of Known Violation

Inheritance: If ϕ entails ψ, then *should* ϕ entails *should* ψ.

The Puzzle of Known Violation (Åqvist 1967):

- **Known Violation**: Max breaks a school rule by eating in the hallway. He should not have been doing that—he should not have been breaking a school rule. But, Max complains, he did not know that he was breaking a rule: he did not know that he was not allowed to eat in the hallway. Still, Max *should have known* the rules, and so he should have known that he was breaking one of them.
- Since knowledge is factive:

  Max knew that he was breaking a school rule $\Rightarrow$ *Max was breaking a school rule.*
- So Inheritance implies that (1) entails (2):

  (1) Max should have known that he was breaking a school rule.
  should (Max knew he was breaking a rule)

  (2) Max should have been breaking a school rule.
  should (Max was breaking a rule)
- But (1) is true while (2) is false.

Professor Procrastinate (Jackson & Pargetter 1986):

- **Professor Procrastinate**: Professor Procrastinate is invited to review a book. He is the only fully qualified expert on the relevant topic and so the best possible outcome to this scenario is the Procrastinate accepts the invitation and writes the review. But Procrastinate procrastinates and he has a terrible time completing projects he undertakes. In particular, if Procrastinate accepts the invitation to review the book, it is extremely unlikely that he will complete the review. This is the worst possible outcome. If, on the other hand, Procrastinate declines, someone else will write the review. The alternative reviewer is less qualified, but will do a serviceable job.
- J & P argue that (3) is true while (4) is false:

  (3) Professor Procrastinate should accept the invitation and write a review.
  should (Procrastinate accepts and writes)

  (4) Professor Procrastinate should accept the invitation.
  should (Procrastinate accepts)

Contextualist treatment of Procrastinate (von Fintel 2012; Blumberg & Hawthorne 2023):

- Exhibit A: When we work to hold the context fixed, it is difficult to affirm (3) while denying (4):

  (5) ? Professor Procrastinate should accept the invitation and write the review, but it's false that he should accept the invitation.
- Exhibit B: Consider the oddity of the following exchange:

  (6) (a) *A*: Professor Procrastinate should accept the invitation and write the review.

(b) B: So he should accept the invitation?
(c) A: ? No, no, I didn't say that.

Contextualist treatment of Known Violation?

- Contrast #1:

  (7) Max should have known that he was breaking a school rule, but he shouldn't have been breaking a rule in the first place.[1]

- Contrast #2:

  (8) (a) A: Max should have known he was breaking a rule.
  (b) B: So he should have been breaking a rule?
  (c) A: No, no, of course not

- Tentative conclusion: Known Violation is a genuine counterexample to Inheritance.

Knowledge and presupposition:

(9) (a) Amy knows that it's raining.
(b) Amy doesn't know that it's raining.
(c) If Amy knows that it's raining, she will bring an umbrella.
(d) Amy might/must know that it's raining.
(e) Does Amy know that it's raining?

Should and presuppositions:

- *Gina stopped plagiarizing* entails and presupposes *Gina plagiarized in the past.*

  (10) (a) Gina didn't stop plagiarizing.
  (b) If Gina stopped plagiarizing, her friends will be pleased.
  (c) Gina probably stopped plagiarizing
  (d) Did Gina stop plagiarizing?

- And note that (11a) does not seem to entail (11b):

  (11) (a) Gina should have stopped plagiarizing.
  (b) Gina should have plagiarized in the past.

- Good Samaritan Paradox (Prior 1958):

  (12) Jone should help Smith who has been robbed, but Smith should not have been robbed.

- **Presupposition Defeats Inheritance**: For many ϕ and ψ such that ϕ entails and presupposes ψ, *should* ϕ does not entail *should* ψ.

[1] I assume that Max is not in a normative dilemma, so that if he should not have been breaking a rule, it's false that he should have been breaking a rule.

2. The meaning of *should*

Standard semantics:

- A sentence of the form *should* ϕ is true at a context iff ϕ is true at all of the best worlds in the modal base of the context (Kratzer 1981).
- A sentence ϕ entails a sentence ψ just in case: any context in which ϕ is true is a context in which ψ is true.
- Inheritance follows from this approach.

"<u>The</u> modal base of context c":

- Imagine there was a context c such that:
  - (1) is true at c just in case: in all of the best worlds in M_1, Max knew he was breaking a rule.
  - (2) is true at c just in case: in all of the best worlds in M_2, Max was breaking a rule.
- Then (1) need not entail (2). But we still want (3) to entail (4), so we need some constraints.

Two constraints on modal bases:

- A modal base M satisfies *should* ϕ's *presupposition condition* iff all of ϕ's presuppositions are true at every world in M.
- A modal base M satisfies *should* ϕ's *diversity condition* iff M contains worlds at which ϕ is true and worlds at which ϕ is false.
- A modal base M is *good* for *should* ϕ iff M satisfies *should* ϕ's presupposition condition and *should* ϕ's diversity condition.

Admissible contexts:

- A context c is *admissible* for an inference of the form *should* ϕ $\Rightarrow$ *should* ψ iff

  c assigns the same modal base to *should* ϕ and *should* ψ, unless there is no modal that is good for both sentences.
- Entailment: ϕ entails ψ iff: for any context c, if c is admissible for the inference from ϕ to ψ, then if ϕ is true at c, ψ is true at c.

Why (1) does not entail (2):

- No modal base is good for both (1) and (2): any modal base that satisfies (1)'s presupposition condition will violate (2)'s diversity condition.
- So no admissible context for the inference from (1) to (2) will assign the same modal base to (1) and (2).
- We can thus find an admissible context at which (1) is true and (2) is false.

Why (3) entails (4):

- There are modal bases that are good for both (3) and (4).
- Example: M contains only two worlds, w and w'.

  In w, Procrastinate accepts and writes.

  In w', Procrastinate does not accept.
- Thus, any admissible context c will assign the same modal base to (3) and to (4). So if (3) is true at c, (4) will be true at c.

3. The norm of assertion

Knowledge Account of assertion (Williamson 1996):

- Wide Norm: One must (assert p only if one knows p).
  $\mathcal{O}(Ap \to Kp)$
  $\neg\mathcal{P}(Ap \wedge \neg Kp)$
- Biconditional Norm: One is epistemically permitted to assert p iff one knows p.
  $\mathcal{P}Ap \leftrightarrow Kp$
  Necessity Norm: $\mathcal{P}Ap \to Kp$
  Sufficiency Norm: $Kp \to \mathcal{P}Ap$

Known Violation: For some q, one knows that (one asserts q and one does not know q). $K(Aq \wedge \neg Kq)$

CLAIM: The Wide Norm, the Biconditional Norm, and Known Violation are jointly inconsistent.

Rosencranz-style argument for CLAIM (Rosencranz 2023):
Let $\phi := (Aq \wedge \neg Kq)$

(i) $K\phi$ (Known Violation)
(ii) $\mathcal{P}A\phi$ (from (i) and Sufficiency)
(iii) $\mathcal{O}(A\phi \to K\phi)$ (from Wide)
(iv) $\mathcal{P}K\phi$ (from (ii), (iii), and an instance of $(\mathcal{O}(\phi \to \psi) \wedge \mathcal{P}\phi) \to \mathcal{P}\psi$)
(v) $\mathcal{P}\phi$ (from (iv) via factivity of K)
(vi) $\mathcal{P}(Aq \wedge \neg Kq)$ (from (v), substituting $(Aq \wedge \neg Kq)$ for ϕ)
(vii) $\neg\mathcal{P}(Aq \wedge \neg Kq)$ (from Wide)

We should reject the step from (iv) to (v):

(13) (a) Max is permitted to know that Mary is breaking a rule.
(b) Mary is permitted to break a rule.

(14) (a) Amy is permitted to know that: she asserted q without knowing q.
(b) Amy is permitted to: assert q without knowing q.

Hoist by his own petard?

- Wide Justification Norm: One must (assert p only if one is justified in believing p).
  $\mathcal{O}(Ap \to Jp)$
  $\neg\mathcal{P}(Ap \wedge \neg Jp)$
- Permissibly Known Violation: for some q, one is permitted to know that: one asserted q and one is not justified in believing q. $\mathcal{P}K(Aq \wedge \neg Jq)$

CLAIM: The Wide Justification Norm and Permissibly Known Violation are jointly inconsistent.

Let $\phi := (Aq \wedge \neg Jq)$

(i) $\mathcal{P}K\phi$ (Permissibly Known Violation)
(ii) $\mathcal{P}\phi$ (from (i) via factivity of K)
(iii) $\mathcal{P}(Aq \wedge \neg Jq)$ (from (ii), substituting $(Aq \wedge \neg Jq)$ for ϕ)
(iv) $\neg\mathcal{P}(Aq \wedge \neg Kq)$ (from Wide Justification Norm)

4. A trivalent semantics for deontic modals and presuppositions

Definition 1. The *sentences of* $\mathcal{L}$ are defined as follows:

(i) Any atomic sentence is a sentence of $\mathcal{L}$.

(ii) If ϕ and ψ are sentences of $\mathcal{L}$ so are: $\neg\phi$, $(\phi \wedge \psi)$, $\mathit{should}_i\ \phi$ (for any $i \in \mathbb{N}$), $(\phi)_\psi$, and $K\phi$.

Assume a fixed model, consisting of a set of possible worlds $\mathcal{W}$, a reflexive binary relation $\mathcal{K}$ on W, and an interpretation function I that maps pairs (p, w) consisting of an atomic sentence of $\mathcal{L}$ and a world in W sinto $\{0,\ 1\}$. Assume that for each propositional valuation v, there is a world $w \in W$, such that $I(w, p) = v(p)$, for all atomics p.

Definition 2. A *variable assignment* is a function R that maps each variable in $\mathbb{N}$ to a serial binary relation on W.

Definition 3. A *context* is a triple $(w, R, \succeq)$ consisting of a world $w \in W$, a variable assignment R, and a total preorder $\succeq$ on W.

Definition 4. We define an evaluation function $[\![\cdot]\!]$ that maps a sentence ϕ and a context into a set of three truth-values $\{1, 0, \#\}$ as follows:

(i) $[\![p]\!]^{w,R,\succeq} \ = \ I(p, w) \quad$ for atomic sentences p

(ii) $$[\![\neg\phi]\!]^{w,R,\succeq} \ = \ \begin{cases} 1 & \text{iff } [\![\phi]\!]^{w,R,\succeq} = 0, \\ 0 & \text{iff } [\![\phi]\!]^{w,R,\succeq} = 1, \\ \# & \text{iff } [\![\phi]\!]^{w,R,\succeq} = \#. \end{cases}$$

(iii) $$[\![(\phi \wedge \psi)]\!]^{w,R,\succeq} \ = \ \begin{cases} 1 & \text{iff } [\![\phi]\!]^{w,R,\succeq} = 1 \text{ and } [\![\psi]\!]^{w,R,\succeq} = 1, \\ 0 & \text{iff } [\![\phi]\!]^{w,R,\succeq} = 0 \text{ or } [\![\psi]\!]^{w,R,\succeq} = 0, \\ \# & \text{otherwise.} \end{cases}$$

(iv) $$[\![(\phi)_\psi]\!]^{w,R,\succeq} \ = \ \begin{cases} 1 & \text{iff } [\![\psi]\!]^{w,R,\succeq} = 1 \text{ and } [\![\phi]\!]^{w,R,\succeq} = 1, \\ 0 & \text{iff } [\![\psi]\!]^{w,R,\succeq} = 1 \text{ and } [\![\phi]\!]^{w,R,\succeq} = 0, \\ \# & \text{otherwise.} \end{cases}$$

(v) $$[\![K\phi]\!]^{w,R,\succeq} \ = \ \begin{cases} 1 & \text{iff } [\![\phi]\!]^{w,R,\succeq} = 1 \text{ and for all } w' \in \mathcal{K}(w), [\![\phi]\!]^{w',R,\succeq} = 1, \\ 0 & \text{iff } [\![\phi]\!]^{w,R,\succeq} = 1 \text{ and there is some } w' \in \mathcal{K}(w), [\![\phi]\!]^{w',R,\succeq} = 0, \\ \# & \text{otherwise.} \end{cases}$$

(vi) $$[\![\mathit{should}_i\ \phi]\!]^{w,R,\succeq} \ = \ \begin{cases} 1 & \text{iff } \textsc{fit}(w, R, \succeq, i, \phi) \ \&\ \forall w' \in \textsc{top}(R_i(w), \succeq), [\![\phi]\!]^{w',R,\succeq} = 1, \\ 0 & \text{iff } \textsc{fit}(w, R, \succeq, i, \phi) \ \&\ \exists w' \in \textsc{top}(R_i(w), \succeq), [\![\phi]\!]^{w',R,\succeq} = 0, \\ \# & \text{otherwise} \end{cases}$$

Definition 5. $\textsc{fit}(w, R, \succeq, i, \phi)$, iff:

(i) for all $w' \in R_i(w)$, $[\![\phi]\!]^{w',R,\succeq} \neq \#$, and

(ii) for some $w' \in R_i(w)$, $[\![\phi]\!]^{w',R,\succeq} = 1$ and
for some $w'' \in R_i(w)$, $[\![\phi]\!]^{w'',R,\succeq} = 0$.

Definition 6. Sentence ϕ *presupposes* ψ iff: for any context $(w, R, \succeq)$, if $[\![\psi]\!]^{w,R,\succeq} \neq 1$, then $[\![\phi]\!]^{w,R,\succeq} = \#$.

Definition 7. Given a set of sentences Γ,

(i) $\mathrm{var}(\Gamma) := \{i \in \mathbb{N} : i \text{ occurs in some sentence in } \Gamma\}$, and

(ii) $R[\mathrm{var}(\Gamma)] := \{R_i : i \in \mathrm{var}(\Gamma)\}$.

Definition 8. We say that a context $(w, R, \succeq)$ is *good* for a set of sentences Γ iff every sentence in Γ is defined at $(w, R, \succeq)$, i.e.

for all $\phi \in \Gamma$, $[\![\phi]\!]^{w,R,\succeq} \neq \#$.

Definition 9. A context $(w, R, \succeq)$ is *admissible* for a set of sentences Γ iff:

(i) $(w, R, \succeq)$ is good for Γ,

(ii) there is no variable assignment R' such that

(a) $(w, R', \succeq)$ is good for Γ, and

(b) $\left|R'[\mathrm{var}(\Gamma)]\right| < \left|R[\mathrm{var}(\Gamma)]\right|$.

Definition 10. A set of sentences Γ *entails* a sentence ϕ iff: for any context $(w, R, \succeq)$, if $(w, R, \succeq)$ is admissible for $\Gamma \cup \{\phi\}$ and $[\![\psi]\!]^{w,R,\succeq} = 1$ for all $\psi \in \Gamma$, then $[\![\phi]\!]^{w,R,\succeq} = 1$.

Fact 1. $K\phi$ entails ϕ, but *should*$_i$ Kp does not entail *should*$_j$ p.

Fact 2. *should*$_i$ $(p \wedge q)$ entails *should*$_j$ p.

Theorem 1. Suppose that neither ϕ nor ψ contains *should* and that ϕ entails ψ. Then *should*$_i$ ϕ entails *should*$_j$ ψ if there is a set of worlds S such that for any R and $\succeq$:

(i) for all $w \in S$, $[\![\phi]\!]^{w,R,\succeq} \neq \#$ and $[\![\psi]\!]^{w,R,\succeq} \neq \#$,

(ii) for some $w \in S$, $[\![\phi]\!]^{w,R,\succeq} = 1$ and for some $w \in S$, $[\![\phi]\!]^{w,R,\succeq} = 0$, and

(iii) for some $w \in S$, $[\![\psi]\!]^{w,R,\succeq} = 1$ and for some $w \in S$, $[\![\psi]\!]^{w,R,\succeq} = 0$.